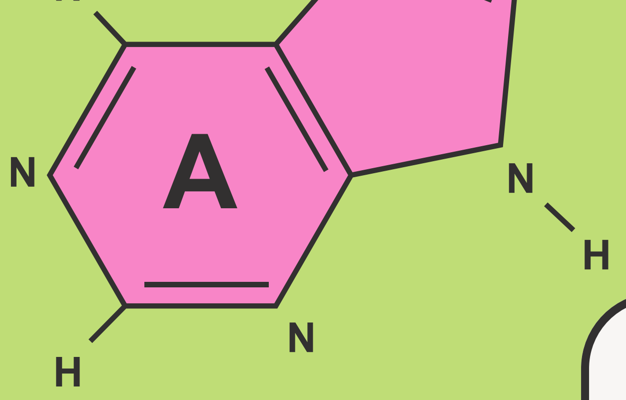
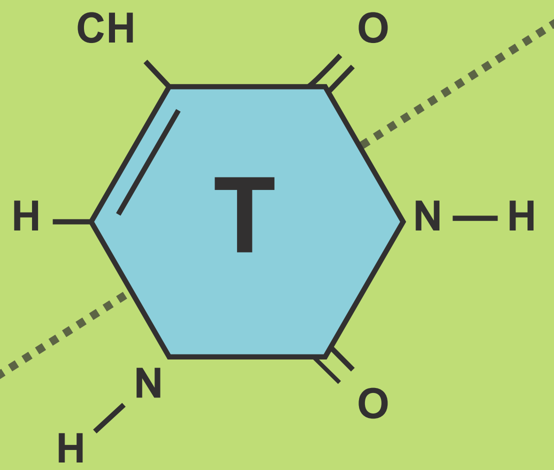
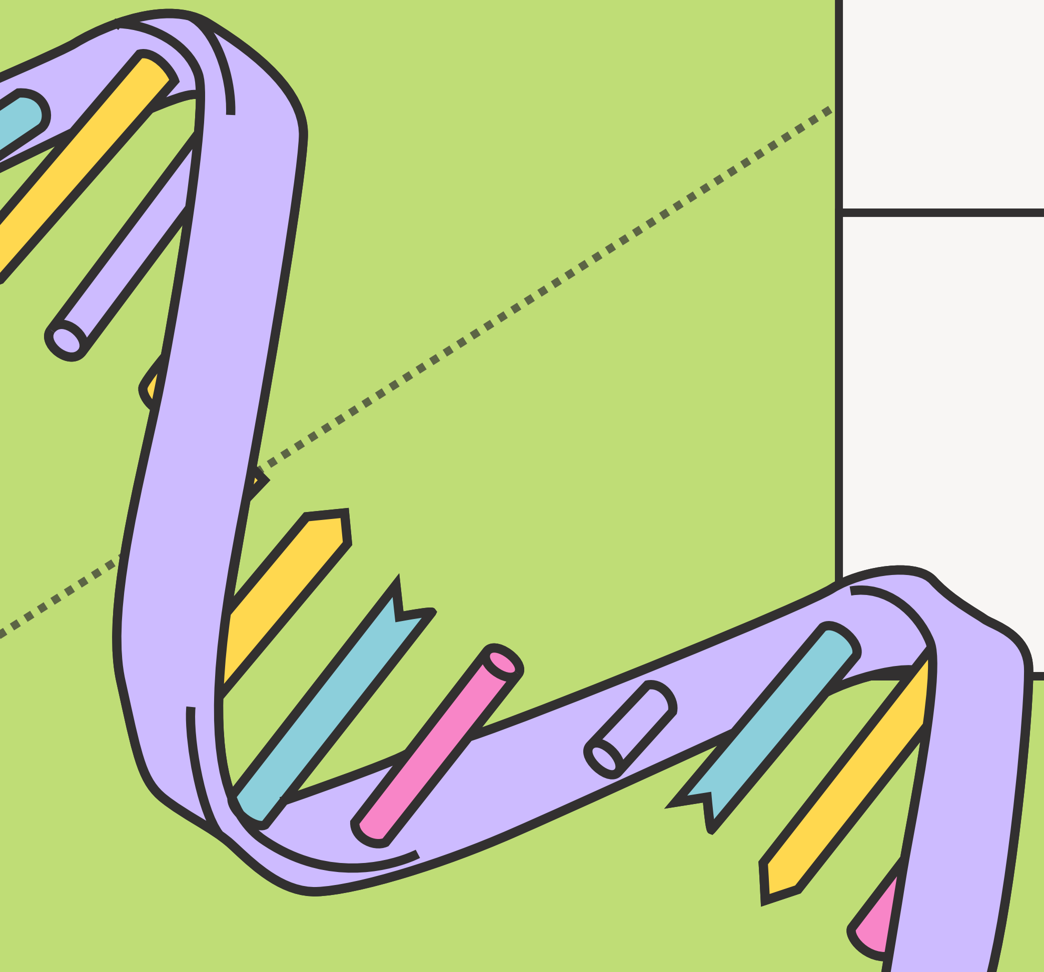


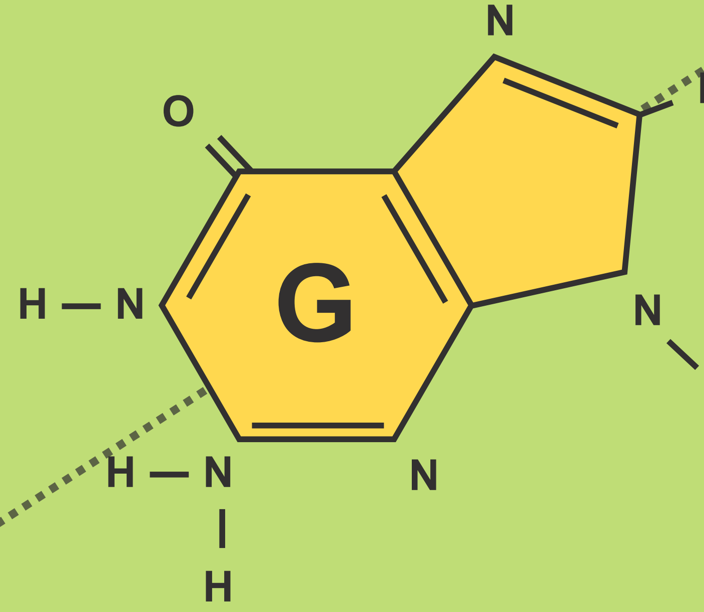


DNA LANGUAGE MODELS FOR MUTATION CLASSIFICATION:

Identifying Benign and Harmful Variants

Presented by: ARJUN SAGAR N V





AGENDA

1. INTRODUCTION
2. DNA AS A LANGUAGE MODEL- THE FOUNDATION
3. TRADITIONAL TOOLS- THE FIRST GEN OF PREDICTORS
4. LANGUAGE MODELS
5. PROBLEM FORMULATION
5. IMPLEMENTATION USING GPN MSA, CNN AND ENSEMBLE MODELS

INTRODUCTION

- DNA is like a language composed of four chemical "letters":
  - Adenine (A), Cytosine (C), Guanine (G), Thymine (T)

- Letters form genetic instructions influencing physical traits (eye color, height) and responses to diseases.

- Mutations are DNA "typos"—small changes that can significantly alter genetic messages.

- **Types of Mutations:**
  - Benign Variants: Harmless; no impact on health.
  - Pathogenic Variants: Harmful; disrupt normal gene functions, causing diseases like cancer, heart disease, or rare genetic disorders.

- Why It Matters: Identifying harmful mutations helps predict, diagnose, and manage genetic diseases, critical for precision medicine and personalized care.

A) WHY MUTATION CLASSIFICATION MATTERS

• Accurate mutation classification directly impacts clinical decisions, patient care, and health outcomes.

• **Clinical Scenario:**
When sequencing a patient's genome, clinicians must determine if identified mutations are:
- Pathogenic (harmful): causing illness and requiring immediate intervention.
- Benign (harmless): inconsequential and not affecting health.

• **Clinical Importance:**
Correct classification influences:
- Treatment strategies and medication choices
- Early intervention and proactive disease management
- Prevention of unnecessary treatments or missed diagnoses

• **Critical Impact Areas:**
- Oncology (Cancer treatment)
- Prenatal diagnostics
- Rare genetic diseases

• Getting classification right can be life-changing, leading to timely therapies and significantly improved patient outcomes.

B) LIMITATIONS OF TRADITIONAL APPROACHES

- Traditional mutation classification methods rely on:
  - Rule-based methods
  - Statistical associations
  - Expert-curated databases
  - Conservation scores (sequence comparison across species)
  - Hand-engineered feature extraction

- **Notable Limitations**:
  - Effective mainly for common, previously documented variants.
  - Poor performance with rare or novel mutations.
  - Lack generalizability across diverse populations and different disease contexts.
  - Inconsistent and sometimes inaccurate predictions.

- **Variants of Uncertain Significance (VUS):**
  - Genetic changes whose clinical impact is unclear.
  - Pose significant challenges to personalized medicine.

- **Impact on Patient Care:**
  - Uncertainty complicates diagnosis, prognosis, and treatment decisions.
  - Limits precision and effectiveness of medical interventions.

C) WHY USE LANGUAGE MODELS ON DNA?

- **Language models (GPT, BERT)** transformed natural language processing by:
  - Identifying complex patterns
  - Interpreting context
  - Extracting meaning from large datasets

- **DNA as Text:**
  - DNA sequences resemble textual data, with nucleotides (A, C, G, T) acting as words or sentences.
  - Biological instructions follow specific "grammatical" patterns.

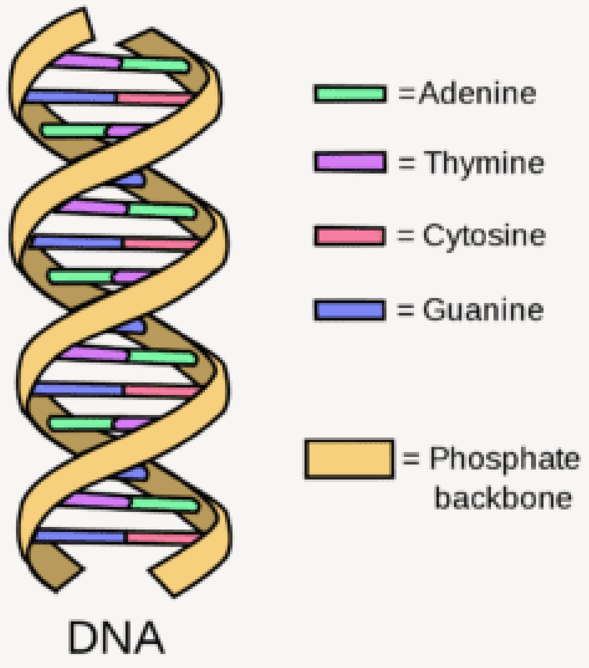
- **Advantages of DNA Language Models:**
  - Detect long-range dependencies missed by traditional methods.
  - Sensitive to subtle genetic variations that affect gene function.
  - Generalize well, effectively classifying unseen or rare mutations without relying on prior documentation

- I**mpact on Mutation Classification:**
  - Faster, more accurate, scalable predictions.
  - Enhanced capability to manage complex genetic diseases.
  - Significantly advances precision medicine and personalized patient care.

DNA AS A LANGUAGE – THE FOUNDATION

- DNA is biological information—a code of instructions fundamental to life itself.

- DNA's Four-letter Alphabet:
  - Adenine (A), Thymine (T), Cytosine (C), Guanine (G)



- Letters form Genetic Words (**Codons**):
  - DNA letters group into triplets called codons.
  - Each codon corresponds to a specific amino acid.

- From **DNA to Proteins**:
  - Chains of amino acids fold into functional proteins, crucial for nearly every biological process.
  - Example: DNA sequence "ATG-GCT-TAC" → amino acids → functional protein within the cell.

- Why "**Language of Life**"?
  - DNA stores, communicates, and executes life's essential instructions.
  - Highlights DNA’s central role in biology and health.

DNA MUTATIONS

Just as typos alter sentences, DNA mutations significantly impact gene function.

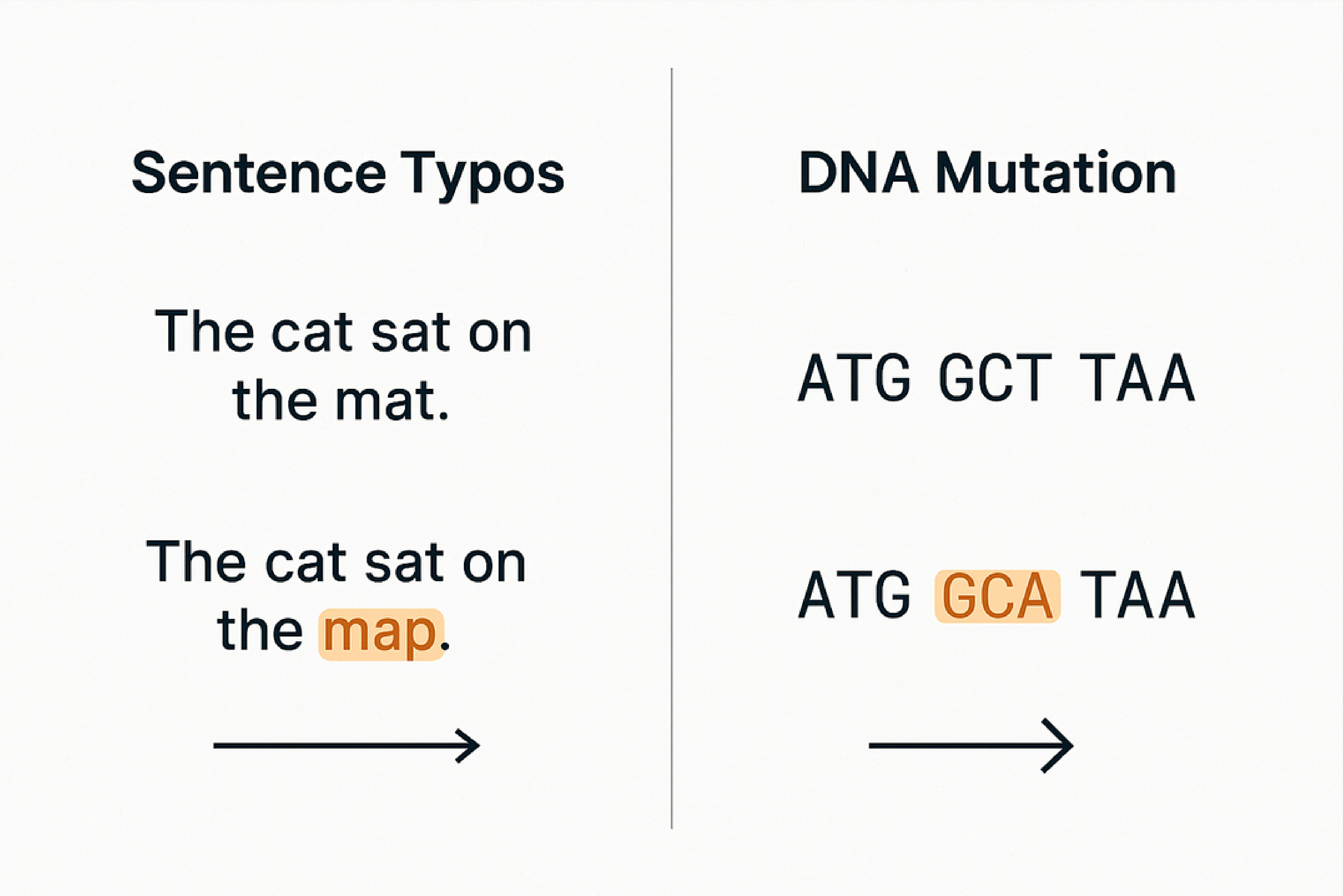
Mutations typically appear as:

- Single Nucleotide Polymorphisms (**SNPs**): Change of a single DNA letter (e.g., A → G).
- Insertions or Deletions (**Indels)**: Addition or removal of DNA letters, altering gene sequences.

Common Mutation Types and Effects:

- **Missense Mutation:** Changes one amino acid, possibly affecting protein function.
- **Nonsense Mutation:** Creates premature stop codon, shortening protein and impairing function.
- **Silent Mutation:** Alters DNA sequence without changing amino acids—often no functional effect.
- **Frameshift Mutation**: Insertion or deletion shifts the DNA reading frame, dramatically changing the protein produced.

The specific type and location of mutations determine their biological impact, highlighting the importance of precise mutation classification in clinical genetics.



Analogy between sentence typos and DNA mutations.

VARIANT CLASSIFICATION – DEFINING HARMFUL VS. BENIGN

When mutations are discovered, classifying their clinical importance involves integrating multiple types of **evidence**:

- Clinical Studies:
- Population Data:
- Functional Experiments:
- Family Studies:

- Reference Databases: Resources like ClinVar and Human Gene Mutation Database (HGMD) compile evidence and clinical interpretations.

Main Classification Categories:

- **Benign**: Harmless; no health impact.
- **Likely Pathogenic**: Potentially harmful; probable link to disease.
- **Pathogenic:** Known disease-causing mutations.
- **Variants of Uncertain Significance (VUS):** Ambiguous mutations with unclear effects.

(Many variants remain uncertain (VUS))

TRADITIONAL TOOLS – THE FIRST GENERATION OF PREDICTORS

Several computational tools have been widely used to classify genetic variants as benign or harmful:

- **SIFT:** Predicts mutations as "Tolerated" or "Damaging" based on sequence conservation.
    - Limitation: Primarily effective for protein-coding regions; struggles elsewhere.

- **PolyPhen-2:** Combines structural and sequence features to classify variants as "Benign" or "Damaging."
    - Limitation: May overfit due to reliance on structural features.

- **CADD:** Integrates multiple annotations and machine learning to provide a numeric pathogenicity score.
    - Limitation: Scores can be challenging for clinical interpretation.

Common Challenges:

- Depend heavily on manually designed features, limiting accuracy for novel or rare variants.
- Predictions can conflict across tools, creating uncertainty for clinicians and researchers.

LANGUAGE MODELS

• In Natural Language Processing (NLP), models like BERT and GPT learn to understand and generate human language.

• These models identify patterns in word sequences, predict missing words, and interpret context.

• Example:
Input – "The cat sat on the ______" → Prediction – "mat"

• Key Concept: These models learn from context and sequence to capture meaning.

• DNA is also a sequence—with four letters (A, T, C, G) instead of 26, and biology as its grammar.

• What if we used NLP models to learn patterns in DNA sequences instead of human language?

ADAPTING LANGUAGE MODELS TO GENOMICS

To adapt language models to genomics, we start by treating DNA like a sequence of words. In NLP, input is made of sentences composed of words. In genomics, we mimic this by breaking DNA into short, overlapping segments called k-mers—these act as our "words."

Example:
DNA sequence: ATGCTGA
3-mers: ATG, TGC, GCT, CTG, TGA

These k-mers form a sentence-like input for the model:
[ATG, TGC, GCT, CTG, TGA]

Just like a model learns the meaning of ambiguous words, it can learn meaningful biological patterns from nucleotide context. These include motifs, splice sites, promoter regions, codon usage, and more.

By training on massive DNA datasets, the model begins to understand the "language of life" — enabling it to predict which mutations may disrupt that language and cause disease.

COMMON TOKENIZATION STRATEGIES

• Tokenization is the first step in applying language models to DNA—breaking sequences into meaningful units called **tokens**.

• **3-mers** are biologically relevant as they mimic codons, which encode amino acids.

• Longer k-mers (e.g., **4-mers to 6-mers**) capture more context and long-range dependencies.

• **Byte Pair Encoding (BPE)**:
- A data-driven method that builds a compact vocabulary by merging frequent k-mer pairs.
- Inspired by subword tokenization in NLP.

• These strategies allow models to:
- Learn directly from raw DNA
- Avoid reliance on handcrafted features
- Enable flexible, scalable, and unbiased genomic understanding

PROBLEM FORMULATION:

Mutation Classification Using GPN-MSA

Goal: Predict whether a DNA variant is Benign or Pathogenic

Core Idea: Use multispecies alignment and a DNA language model (GPN-MSA) to estimate how evolutionarily plausible a mutation is.

Step 1: **Represent the Sequences**

Let:
- **S:** Reference DNA sequence
- **S̃:** Mutated DNA sequence (with SNV)
- **M**: Multispecies alignment context
- **fθ:** GPN-MSA model (pretrained masked DNA language model)

We analyze a small window of sequence centered at the mutation site.

Example:
- Variant is at chr1:100, changing G → A
- Reference: G, Alternate: A, Model: Pretrained GPN-MSA

PROBLEM FORMULATION:

- Step 2: **Extract Sequence Window**

- Choose a window size w (e.g., 3 bases on each side).
- Extract a centered window around the mutation.

Example (w = 3):

- Reference sequence: ACT<u>G</u>GTA
- Alternate sequence: ACT<u>A</u>GTA

- Step 3 – **Use Multispecies Alignment (MSA)**

- Align corresponding DNA sequences from other species.
- These provide evolutionary context for the variant site.

- Example MSA:
    - Human: ACTGGTA
    - Chimpanzee: ACTGGTA
    - Mouse: ACTGGTC
    - Zebrafish: ACTGGAA

PROBLEM FORMULATION:

- Step 4 – **Compute Base Likelihoods**

- GPN-MSA computes how likely a base is at the variant position.
- Estimate:
  - log P_ref = likelihood of original base
  - log P_alt = likelihood of mutated base

- Example Output:
  - log P(G | context) = –0.3
  - log P(A | context) = –1.2

- Step 5 – **Compute Evolutionary Index (EI)**

- Calculate Evolutionary Index:
  - EI = log P_ref – log P_alt

- Interpretation:
  - EI > 0 → Mutation is less likely → Pathogenic
  - EI < 0 → Mutation is more likely → Benign

- Example:
  - EI = –0.3 – (–1.2) = 0.9
  - Interpretation: Mutation is less plausible → Pathogenic

PROBLEM FORMULATION:

- Step 6 – **Final Classification**

Use a threshold τ (e.g., 0.5) to classify:

  - If EI > τ → Predict Pathogenic
  - If EI < τ → Predict Benign

- Example:
  - Threshold τ = 0.5
  - EI = 0.9 > 0.5 → Classify as **Pathogenic**

DATASET

- **clinvar.vcf.gz** – Variant Call Format (VCF) from ClinVar:
  - Contains genomic variants and their clinical significance (e.g., benign, pathogenic).
  - Based on GRCh38 human genome assembly.
  - Each row includes:
    - CHROM, POS, ID, REF, ALT, QUAL, FILTER, and INFO
  - INFO field includes rich annotations:
    - Clinical significance, conditions, gene names, phenotypic effects
  - Useful for:
    - Mutation classification
    - Labeling training data
    - Disease variant analysis

IMPLEMENTATION USING CNN

ONLY FOR REFERENCE SEQUENCE:

Epoch	Loss	Train Accuracy
01	0.5191	79.20%
02	0.4905	79.20%
03	0.4399	79.40%
04	0.3878	81.73%
05	0.3238	84.65%
06	0.2620	88.62%
07	0.1985	92.22%
08	0.1452	94.38%
09	0.1075	96.00%
10	0.0872	96.80%

Table 1: Training progress over 10 epochs

Metric	Class 0 (Benign)	Class 1 (Pathogenic)
Precision	0.8494	0.4891
Recall	0.9128	0.3401
F1-Score	0.8800	0.4012
Support	803	197

Table 2: Classification metrics for each class

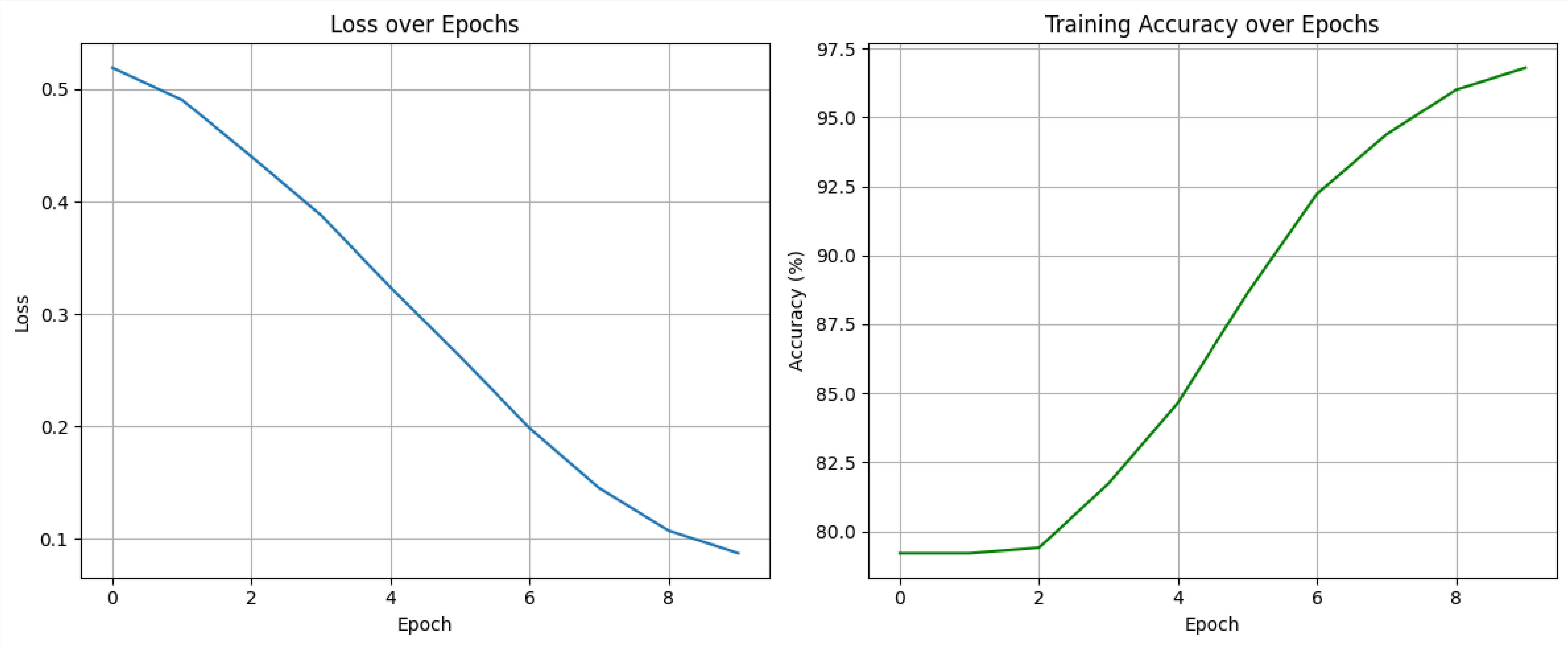
Overall Performance

- **Test Accuracy:** 80.00%
- **ROC AUC Score:** 0.7560
- **Model saved as:** `model.pth`

IMPLEMENTATION USING CNN

Click here for Implementation code:

ONLY FOR REFERENCE SEQUENCE:



Testing on Random DNA sequence:

```
Predicted output:
Sequence:      CACATCGTGCTTCTGGCGTCGTGAACTTCGCGTGCCTCCGCTCGTTTGCAACACGGTTCATTGTCGTGTCCCAGGCGGGCTCAGGCGGGCATCCCATTTAG
Probability:  0.8031
Label:        Pathogenic (1)
```

IMPLEMENTATION USING CNN AND GPN MSA

FOR REFERENCE AND ALTERNATE SEQUENCE:

Epoch	Loss	Test Accuracy	Val AUC
1	0.5231	78.27%	0.6176
2	0.5009	78.27%	0.6544
3	0.4752	78.27%	0.6493
4	0.4259	78.40%	0.6959
5	0.3713	79.73%	0.7421
6	0.3021	78.80%	0.7500
7	0.2443	81.87%	0.7513
8	0.1761	78.80%	0.7634
9	0.1354	80.53%	0.7509
10	0.1098	80.80%	0.7569
11	0.0838	81.33%	0.7635
12	0.0643	81.47%	0.7520
13	0.0523	80.00%	0.7580
14	0.0466	80.27%	0.7568

Table 1: Epoch-wise Training Summary (early stopping after epoch 14)

- Accuracy: **80.80%**
- Macro Avg F1-Score: 0.6690
- ROC AUC Score: **0.7828**
- Model saved to `training_history.pth`

Metric	Class 0	Class 1
Precision	0.8585	0.5263
Recall	0.9100	0.4000
F1-Score	0.8835	0.4545
Support	600	150

Table 2: Final Test Set Evaluation

Metric	Class 0	Class 1
Precision	0.8490	0.6049
Recall	0.9467	0.3267
F1-Score	0.8952	0.4242
Support	600	150

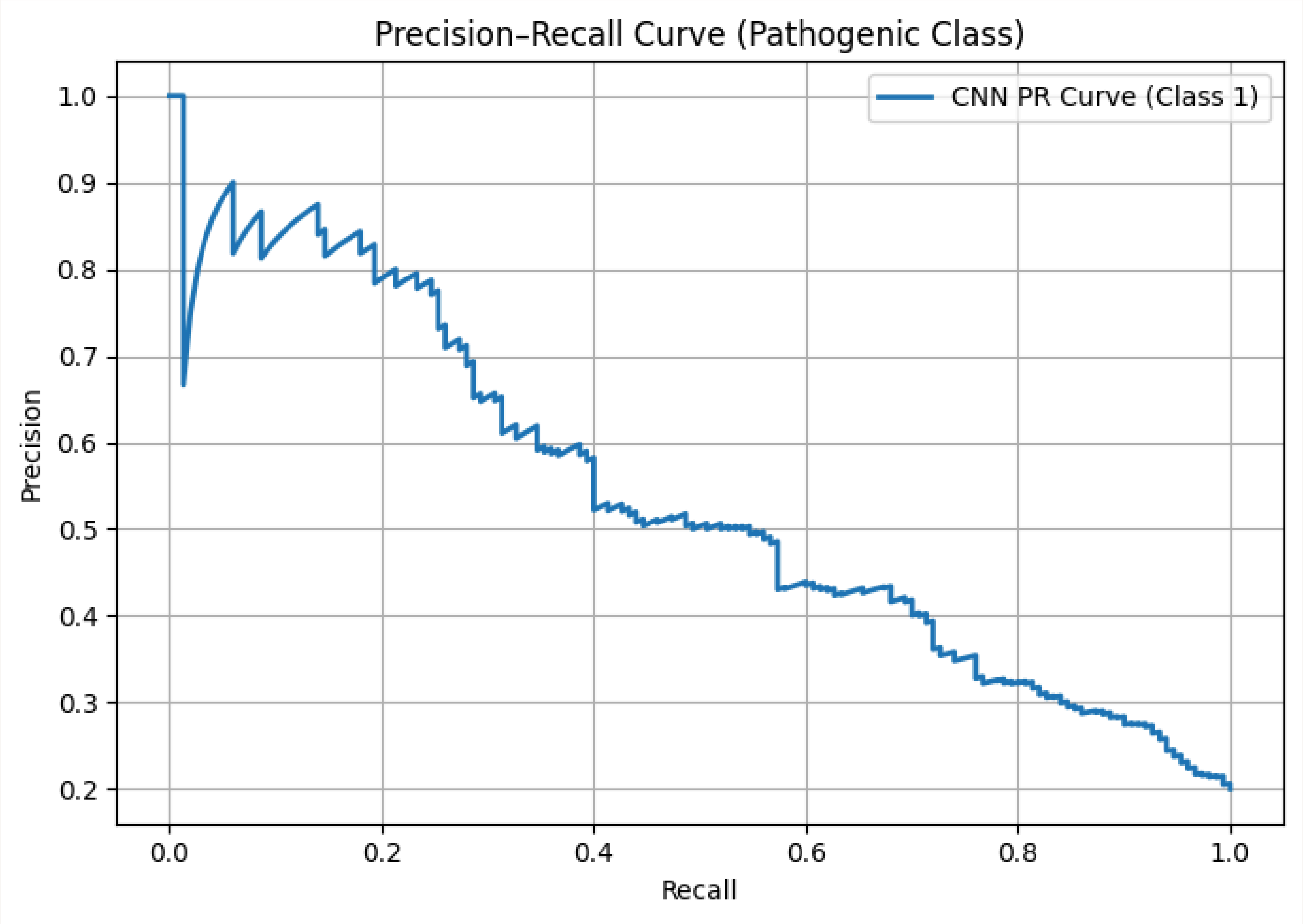
Table 3: Ensemble Model Evaluation

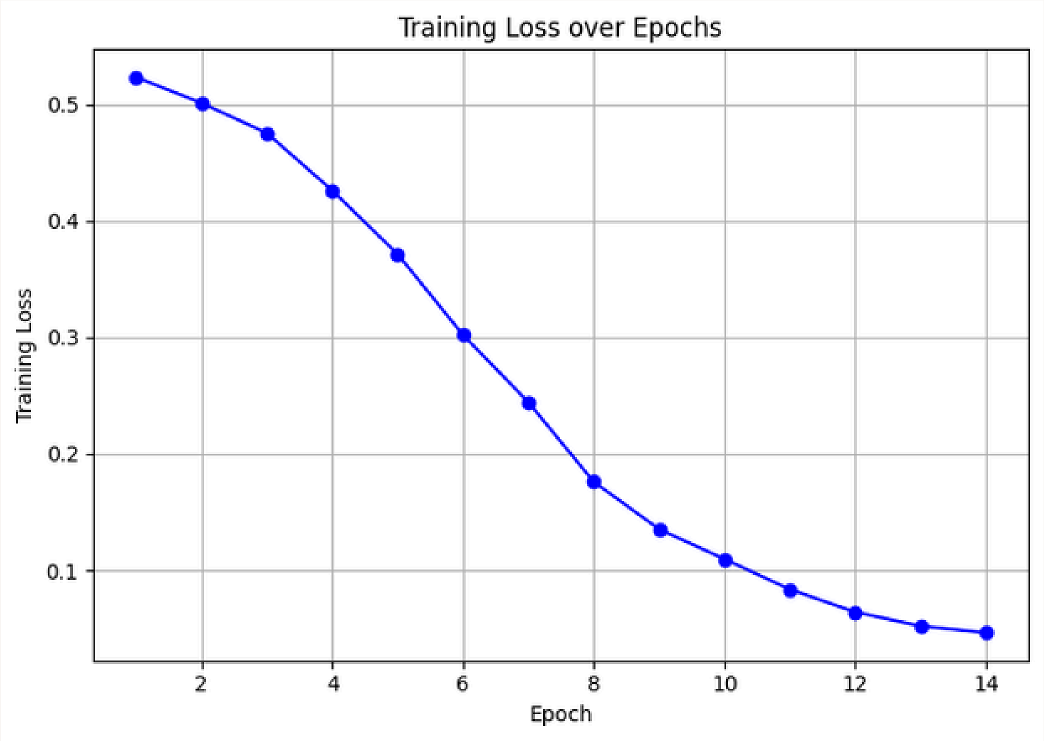
Ensemble Metrics:

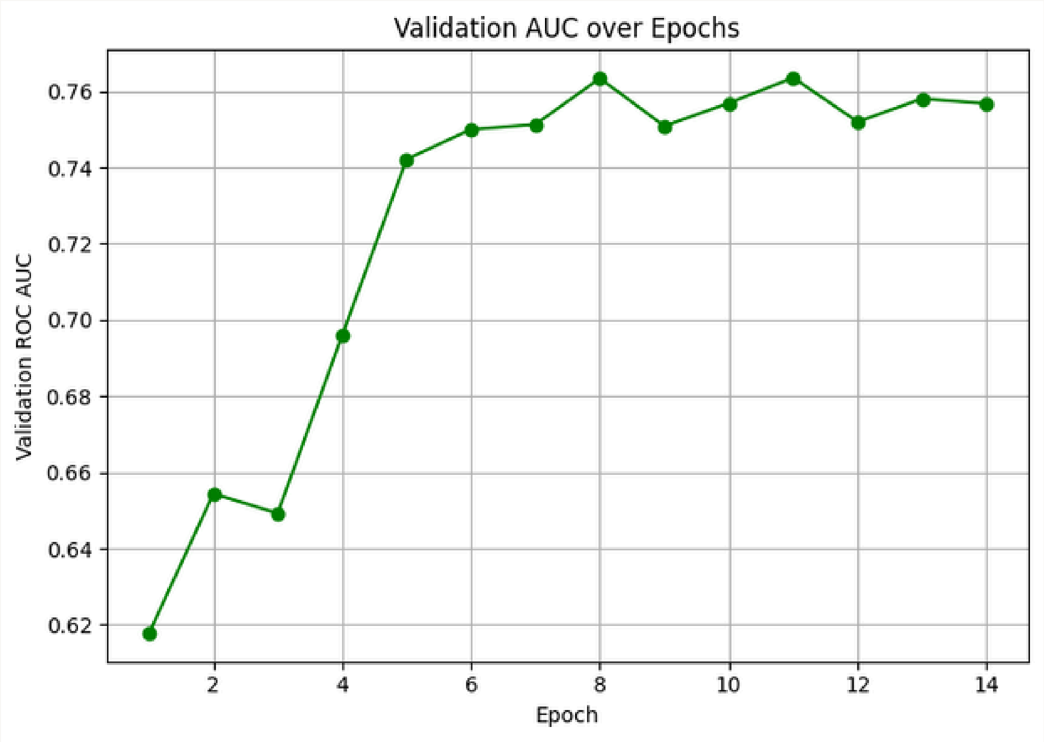
- Accuracy: **82.27%**
- ROC AUC Score: **0.7828**

IMPLEMENTATION USING CNN AND GPN MSA

FOR REFERENCE AND ALTERNATE SEQUENCE:







IMPLEMENTATION USING CNN AND GPN MSA

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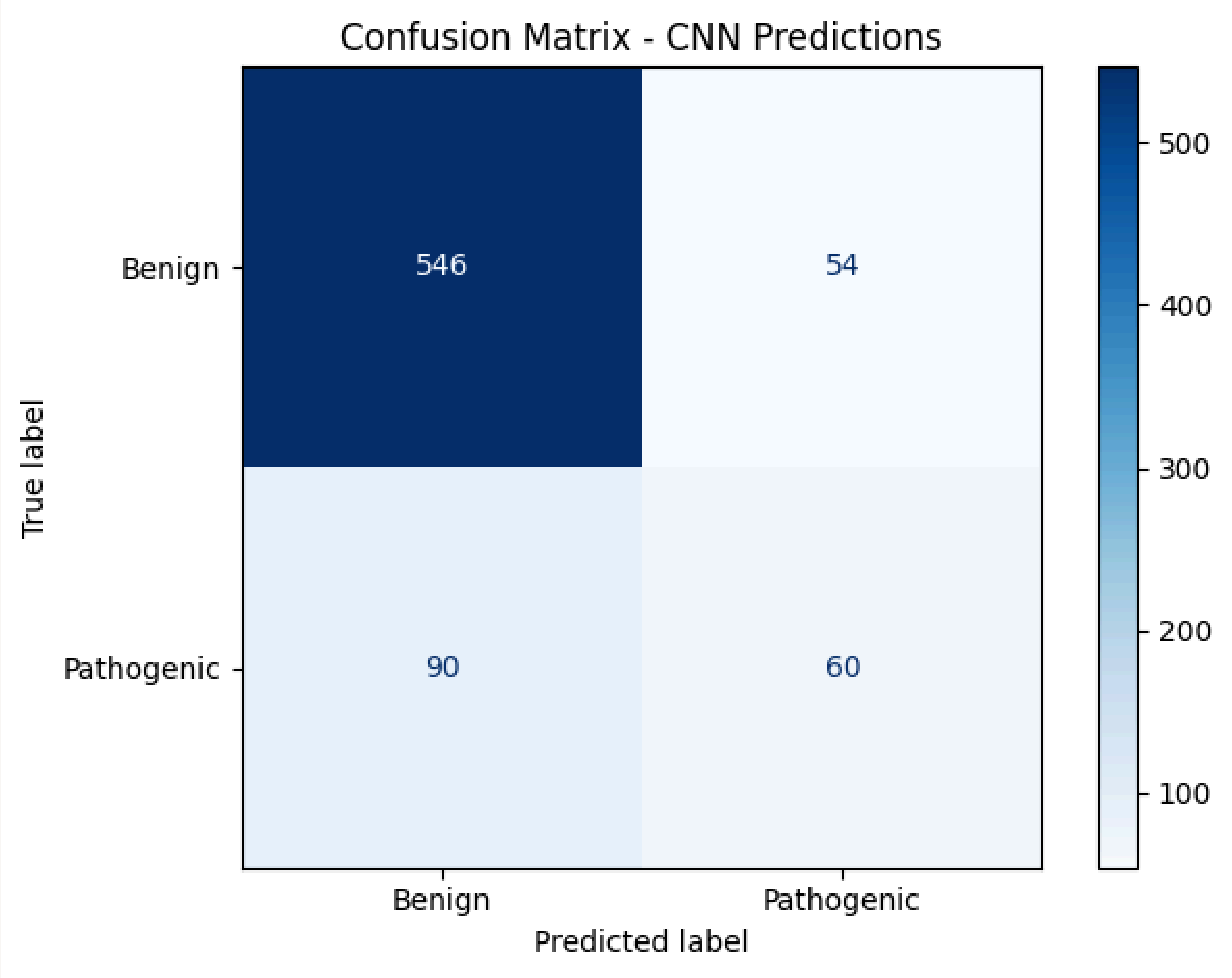
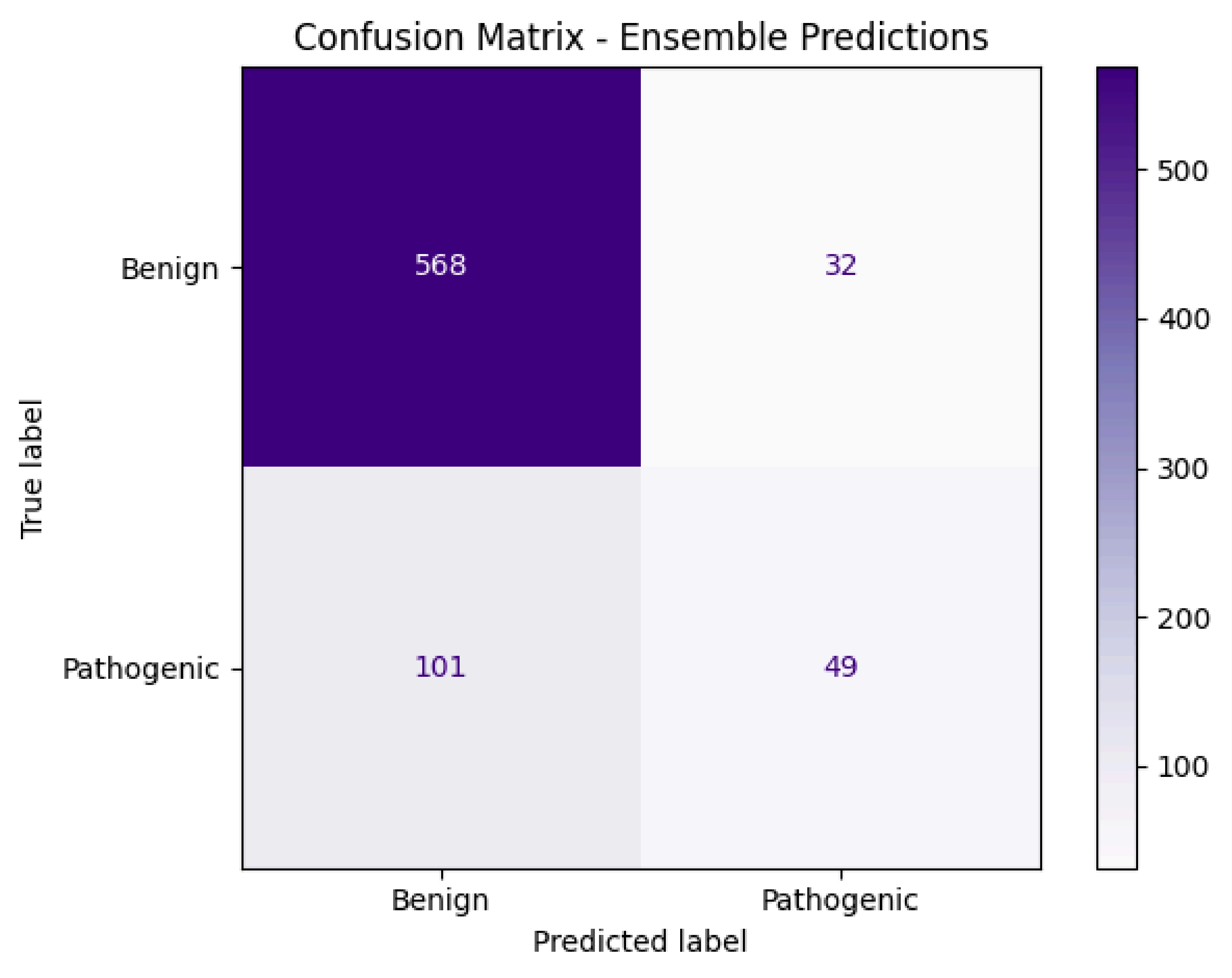
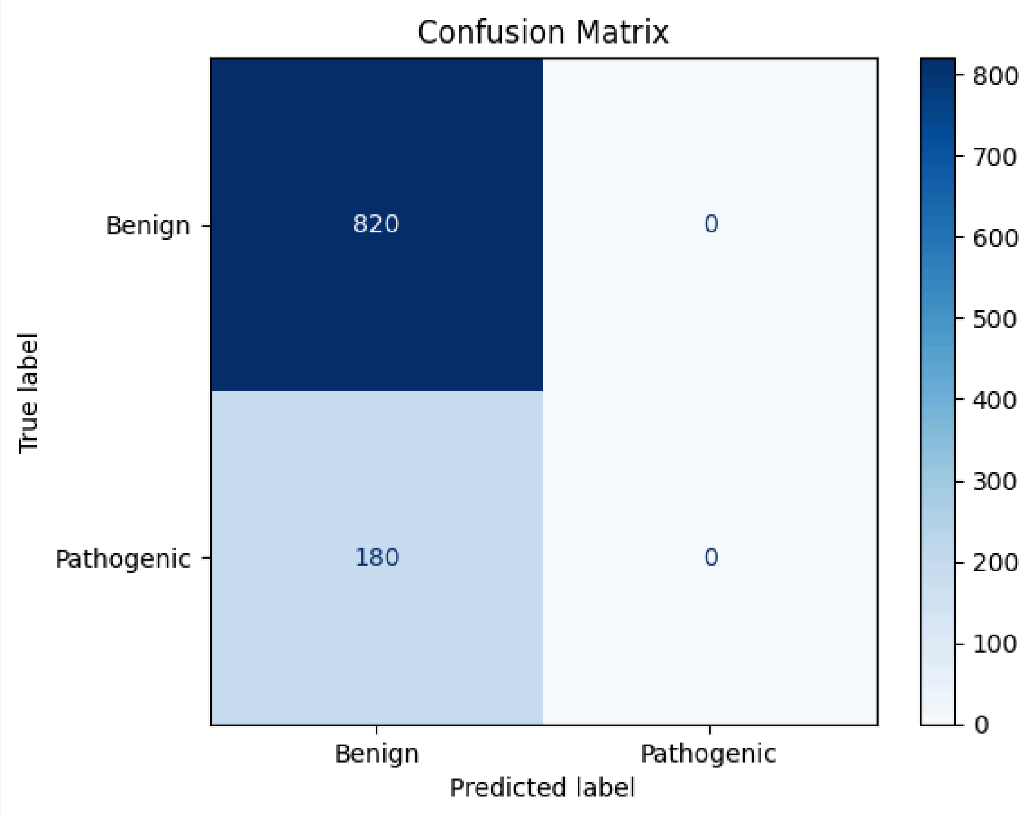




Click here for
Implementation code:

IMPLEMENTATION USING LANGUAGE MODEL

Epoch	MLM Loss		Classification Loss	Accuracy
1	0.2680	→	0.4837	81.67%
2	0.2228	→	0.4757	82.05%
3	0.2214	→	0.4739	82.05%



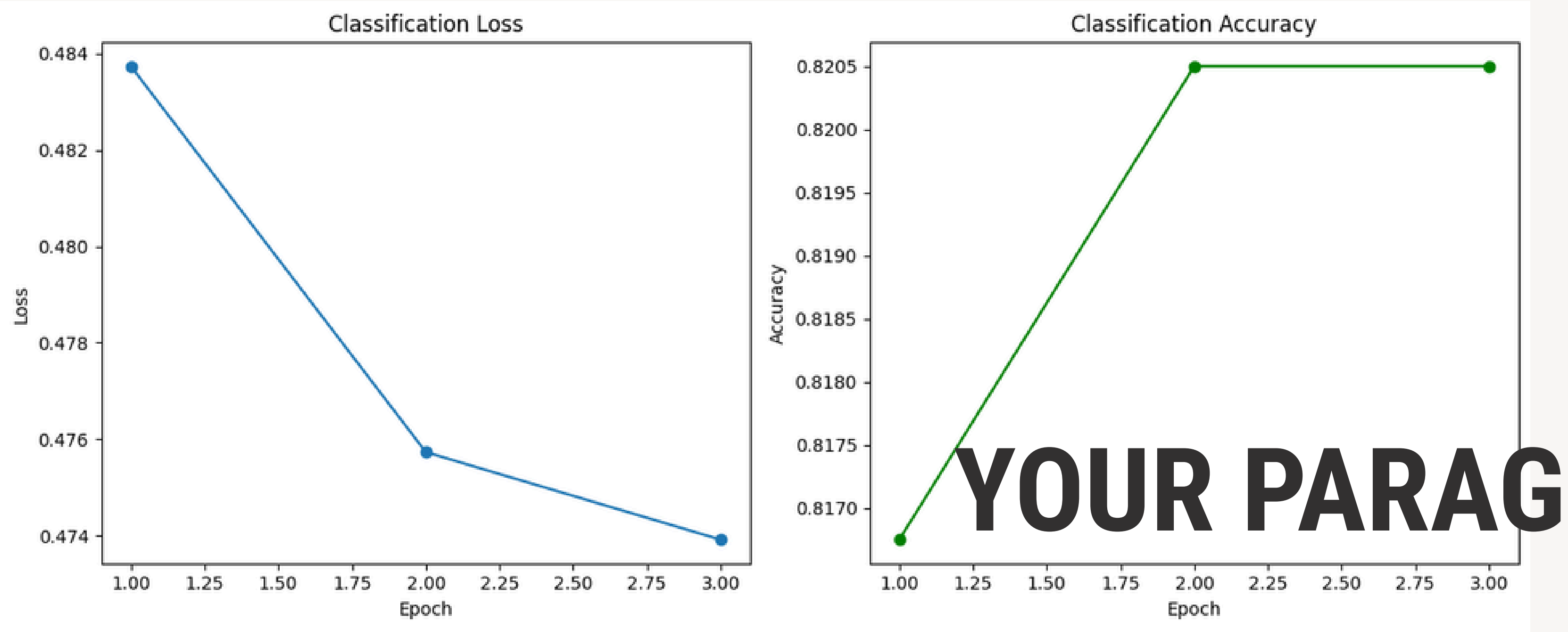

```
Final Accuracy: 0.8200

Classification Report:
              precision    recall  f1-score   support

      Benign       0.82      1.00      0.90       820
  Pathogenic       0.00      0.00      0.00       180

    accuracy                           0.82      1000
   macro avg       0.41      0.50      0.45      1000
weighted avg       0.67      0.82      0.74      1000
```

IMPLEMENTATION USING LANGUAGE MODEL



YOUR PARAGRAPH TEXT

Click here for
<u>Implementation code:</u>

```
#  known sequence from DataFrame
example_sequence = df.iloc[0]["sequence"]

predict_sequence(model, example_sequence, tokenizer)
```

```
Predicted Class: Benign
Probabilities → Benign: 0.8475, Pathogenic: 0.1525
```

Thank you